W-B-177B
July 6, 1976
SUPERSEDING
Fed. Spec. W-B-177A
May 12, 1964

FEDERAL SPECIFICATION

BATTERY FILLER, GRAVITY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers containers to dispense distilled water to the cells of storage batteries.

1.2 Classification.

1.2.1 Types and styles. Battery fillers furnished under this specification shall be of one type (jug) and the following styles, as specified (see 6.2):

Style A - With hose dispenser.
Style B - With valve control pouring spout.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Wood, Cleated, Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging and Packing Levels.
Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specification:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Atributes.
MIL-STD-123 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard product. The equipment shall be essentially the standard product of the manufacturer so that prompt and continuing service and delivery of spare parts may be assured. The component parts of the unit need not be the products of the same manufacturer.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of a good commercial quality entirely suitable for the purpose. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.2.1 The battery fillers shall be molded from a natural or synthetic rubber compound or a thermoplastic material capable of meeting the performance requirements of 3.6.

3.3 Construction.

3.3.1 Style A fillers. Style A fillers shall consist of a semi-closed shell, molded in one or two pieces, with an integral bail-type lifting handle at the top and an eye or buttoned-in holder for holding the nozzle end of the filling hose when not in use.

3.3.1.1 Hose and fittings.

3.3.1.1.1 Hose. The hose shall be attached not more than 1 inch from the bottom of the interior of the shell. The hose shall measure 18 inches ± 3 inches long with an inside diameter of 1/4 inch ± 1/16 inch. The dispensing end of the hose may have a small nonmetallic nozzle to restrict and direct the liquid flow. Such nozzle must not be affected by sulfuric acid or potassium hydroxide found in battery cells.

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3.3.1.1.2 Fittings. Fittings in contact with battery water shall be of such materials as will not contaminate the water to be used for filling batteries containing sulfuric acid or potassium hydroxide.

3.3.2 Style B fillers. Style B fillers shall consist of a completely enclosed shell molded in one piece with an integral pitcher-type lifting handle and an integrally molded spout so positioned at the top as to provide ease of handling during the dispensing operation. The spout shall accommodate a removable control valve assembly which automatically limits the addition of water to the battery cells to prevent overfilling.

3.4 Capacity.

3.4.1 Style A fillers. Style A fillers shall have a minimum capacity of two quarts.

3.4.2 Style B fillers. Style B fillers shall have a minimum capacity of two quarts.

3.5 Weight. The empty battery fillers shall not exceed four pounds in weight.

3.6 Performance.

3.6.1 Gravity drop. The filler shall be subjected to the drop test of 4.4.1 without suffering any apparent injury to any part of the filler. Minor indentations less than 1/8 inch are acceptable.

3.6.2 Freezing. The filler shall be subjected to the test of 4.4.2 without suffering any apparent injury to any part of the filler.

3.6.3 Handle design. The handle shall be so designed that it shall not suffer any apparent injury when the filler is subjected to test of 4.4.3.

3.6.4 Temperature condition. The filler shall be capable of satisfactory performance in any ambient temperature from 125°F with full impact of solar radiation of 360 B.t.u./Ft.Sq./Hr. for at least 4 hours to 32°F for at least 3 days without benefit of solar radiation, and shall not be damaged by storage (at any ambient temperature from 160°F for periods of 4 hours daily to minus 80°F for periods of at least 3 days duration.)

3.7 Identification marking. Each item of equipment shall bear the manufacturer's name or trademark on a nameplate securely affixed in a conspicuous place. In lieu of the nameplate, the manufacturer's name or trademark may be molded integral with, stamped, or otherwise permanently marked upon the item of equipment.

3.8 Finish. The equipment shall be finished in accordance with the manufacturer's standard practice for this type of equipment.

3.9 Workmanship. Workmanship shall be of the highest grade throughout and in accordance with good commercial practice for this type of equipment. The battery filler shall be suitable for the intended purpose and free of defects which may affect its performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Sampling.

4.2.1 Visual and mechanical inspection. Sampling for visual and mechanical inspection shall be in accordance with MIL-STD-105, inspection level II, and acceptable quality levels (AQL) of 1.0 percent for major defects and 4.0 percent for minor defects.

4.2.2 Test. Sampling for test shall be in accordance with MIL-STD-105, inspection level S-4, AQL 4.0 percent.

4.3 Inspection. Battery fillers sampled in accordance with 4.2.1 shall be visually inspected to determine compliance with the requirements of this specification as follows:

Major defects:

Container material (3.2).
Container construction (3.3.1 and 3.3.2).
Hose nozzle material (3.3.1.1.1).
Fittings material (3.3.1.1.2).
Container weight (3.5).

Minor defects:

Hose dimensions (3.3.1.1.1).
Containers capacity (3.4).
Markings (3.7).
Finish (3.8).
Workmanship (3.9).

4.4 Tests. Battery fillers sampled in accordance with 4.2.2 shall be subjected to the following tests to determine compliance with the requirements of 3.6.

4.4.1 Drop test. The filler (and hose when applicable) shall be completely filled with water and subjected to a free fall onto a nonresilent surface from a height of three (3) feet. Each battery filler shall be drop tested three (3) times. The orientation of drop shall be with the base parallel to the impact surface. The temperature of the entire unit shall be 34° ± 2°F.

4.4.2 Freeze test. The filler (and hose when applicable) shall be completely filled with water and be subjected to a temperature of 25° ± 2°F for 24 hours and then warmed slowly to 40° ± 2°F. This shall be repeated three times.

4.4.3 Handle strength test. The filler (and hose when applicable) shall be completely filled with water and be subjected to a free fall of 3 feet from the end of a rope. This shall be repeated three times.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Each battery filler shall be packaged in fiberboard box conforming to PPP-B-636, class weather resistant. Boxes shall be closed and scaled in accordance to the appendix of PPP-B-636.

5.1.2 Level B. Each battery filler shall be packaged in a paperboard or fiberboard box conforming to PPP-B-566, PPP-B-665, PPP-B-676, or PPP-B-636.

5.1.3 Level C. Battery fillers shall be packaged in accordance with manufacturer's standard commercial practice.

5.2 Packing. Packing shall be Level A, B, or C, as specified.

5.2.1 Level A. Packaged battery fillers shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The boxes shall be closed in accordance with method I in the appendix to PPP-B-636.

5.2.3 Level C. Packaged battery fillers shall be packed in containers conforming to the National Motor Freight Classification or the Uniform Freight Classification.

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6. NOTES

6.1 Intended use. The battery filler is intended for use in servicing automotive and truck electric storage batteries. Extreme caution should be exercised to prevent contamination of storage batteries through the use of the battery filler. In cases where a filler is used to service both acid and alkali type batteries all filler parts that come in contact with battery electrolyte should be washed thoroughly after servicing a particular battery type.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Style of filler (see 1.2.1).
(c) Inspection responsibility, if other than specified (see 4.1).
(d) Level of packaging and packing required (see 5.1 and 5.2).
(d) Special marking, if required (see 5.3).

6.3 This specification is believed to adequately describe the item under consideration. Normally no samples will be necessary prior to award to determine specification compliance. When samples of bids are believed necessary, both the request for the sample and the purpose thereby subserved should be specifically stated in the invitation for bids.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

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